

Network Queueing Models

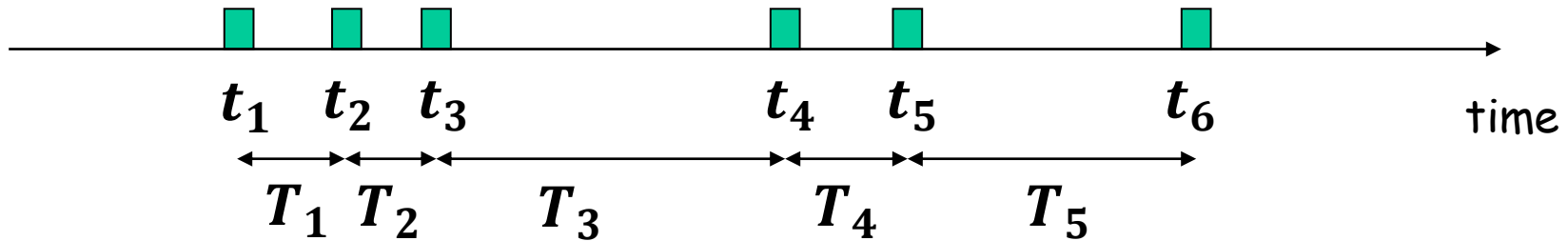
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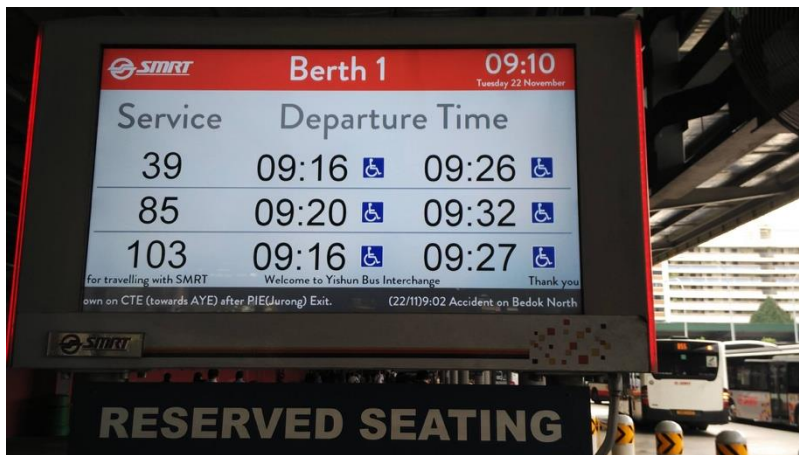
CS4226: Internet Architecture

Model the packet flow: Arrival Pattern



- Arrival time t_i
- Inter-arrival time $T_i \triangleq t_{i+1} - t_i$
- Model the inter-arrival times T_i s by independent and identically distributed (i.i.d.) random variable (r.v.) T

Thinking about buses ...



Random Experiment



- Consider a random experiment whose outcome cannot be determined in advance
- **Sample space S** : the set of all outcomes
- **Event E** : a subset of the sample space
 - ❖ An event E is occurred if the outcome $s \in E$
- **Probability function $P(E)$**
 - ❖ $0 \leq P(E) \leq 1$
 - ❖ $P(S) = 1$
 - ❖ $P(\bigcup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} P(E_i)$ for any sequence of events $E_1, E_2, \dots$ that are *mutually exclusive*

Random Variable

□ A **random variable** X is a function that assigns a real value to each outcome $s \in S$.

□ For any set of real numbers $A \subseteq \mathcal{R}$

$$P\{X \in A\} \stackrel{\text{def}}{=} P(X^{-1}(A))$$

□ The **distribution function** F of the r.v. X is defined on any real number x by

$$F(x) \stackrel{\text{def}}{=} P\{X \leq x\} = P(X^{-1}((-\infty, x]))$$

□ A r.v. is continuous if there exists a **probability density function** $f(x)$ such that

$$f(x) \stackrel{\text{def}}{=} \frac{d}{dx} F(x)$$

Random Variable

- Two random variables are independent if the realization of one does not affect the probability distribution of the other.

$$f_{X,Y}(x, y) = f_X(x)f_Y(y)$$

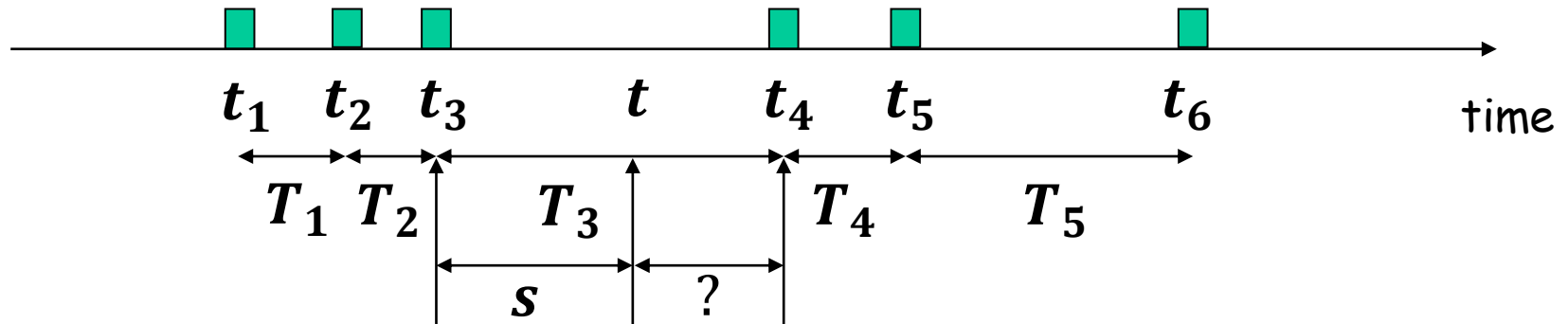
- The expectation or mean of a random variable X

$$E[X] \stackrel{\text{def}}{=} \int_{-\infty}^{+\infty} xf(x) dx \quad \text{or} \quad E[X] \stackrel{\text{def}}{=} \sum_{x=-\infty}^{+\infty} xP\{X = x\}$$

Exponential Distribution

- A continuous r.v. T follows/has an **exponential distribution** with parameter $\lambda > 0$, if, for $x \geq 0$
 - ❖ $F(x) = P\{T \leq x\} = 1 - e^{-\lambda x}$ or $\bar{F}(x) = P\{T > x\} = e^{-\lambda x}$
 - ❖ $f(x) = \frac{dF(x)}{dx} = \lambda e^{-\lambda x}$
- $E[T] = \int_{-\infty}^{\infty} x f(x) dx = \frac{1}{\lambda}$
- What is the unit and physical meaning of λ ?
- ❖ Why is exponential a good distribution?
 - ❖ Memoryless property: $P\{T > s + t \mid T > s\} = P\{T > t\}$

Memoryless Property



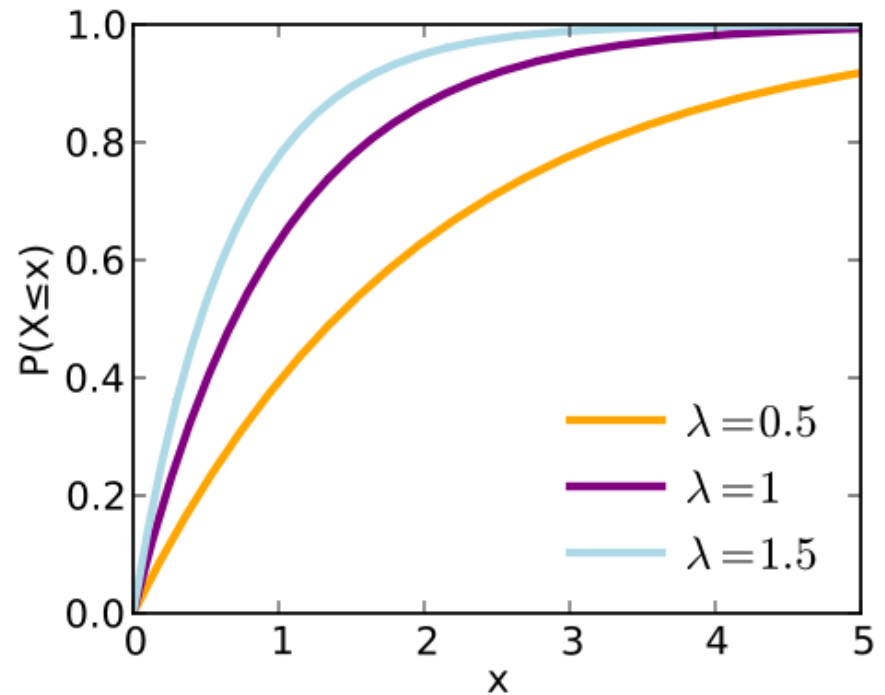
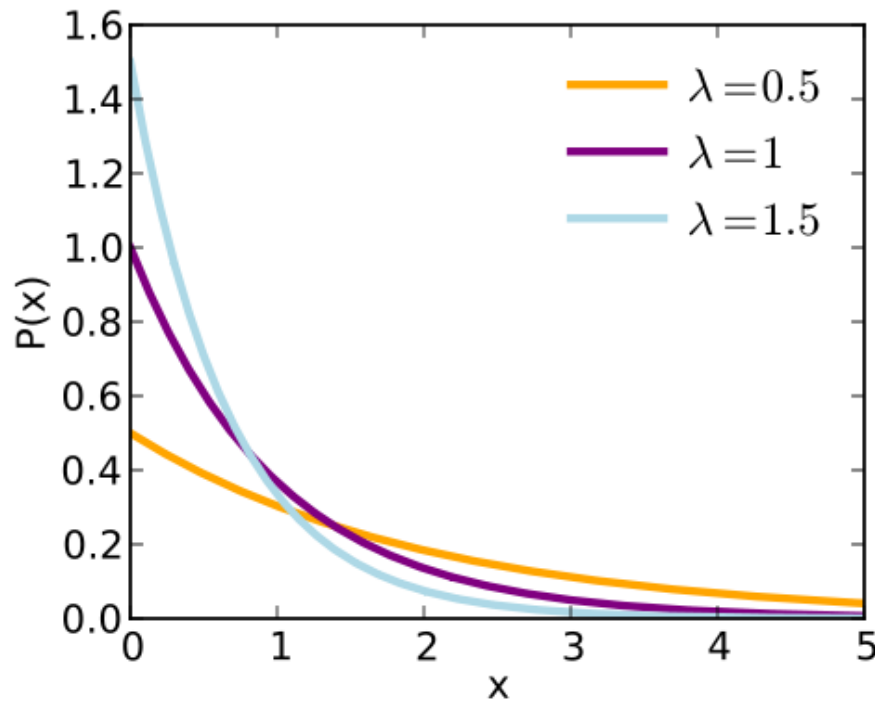
□ What is the distribution of $T_3 - s$?

$$P\{T_3 - s > x \mid T_3 > s\} = P\{T_3 > s + x \mid T_3 > s\}$$

□ If T_3 follows an exponential distribution

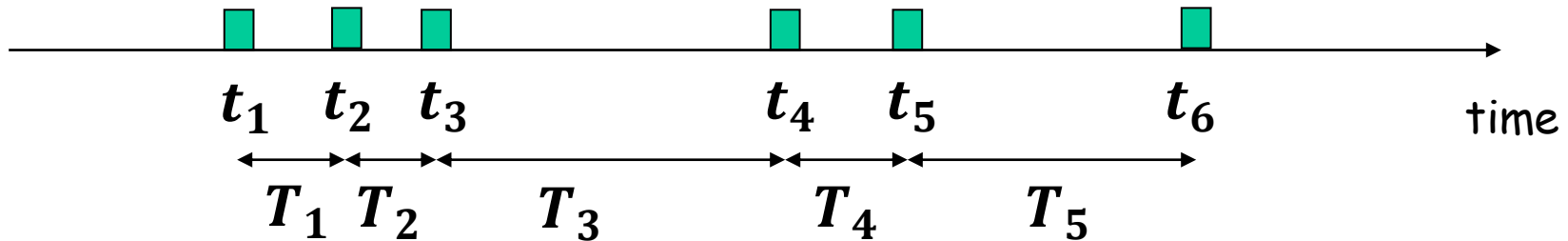
$$\Rightarrow P\{T_3 > s + x \mid T_3 > s\} = P\{T_3 > x\} \text{ for all } x \geq 0$$

Exponential Distribution



From: http://en.wikipedia.org/wiki/Exponential_distribution

Model the packet flow: Poisson Process



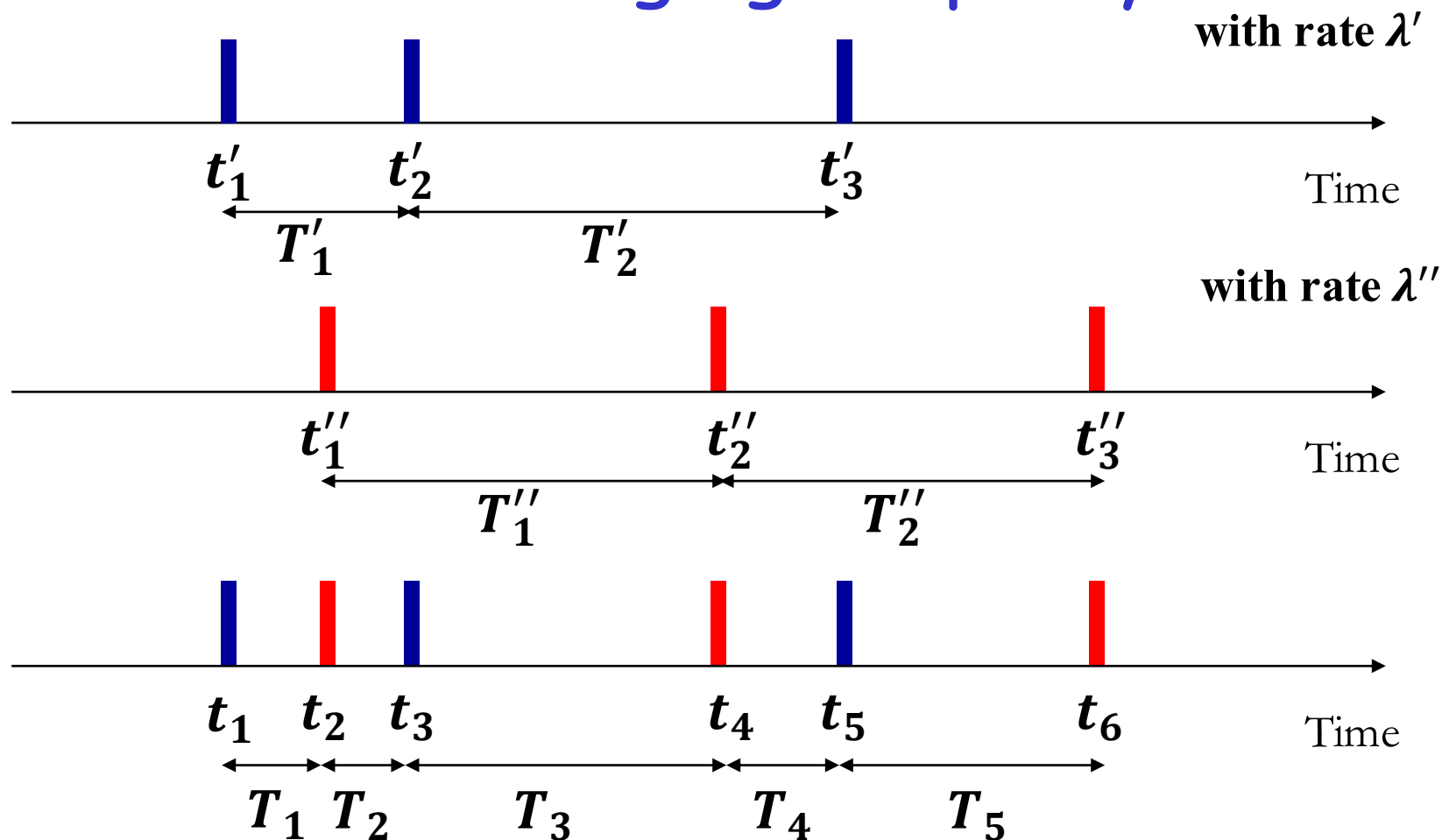
□ T_i s are i.i.d. r.v.s

- ❖ exponentially distributed as T with rate λ
- ❖ this arrival pattern is called a Poisson process

□ Why is Poisson a good model?

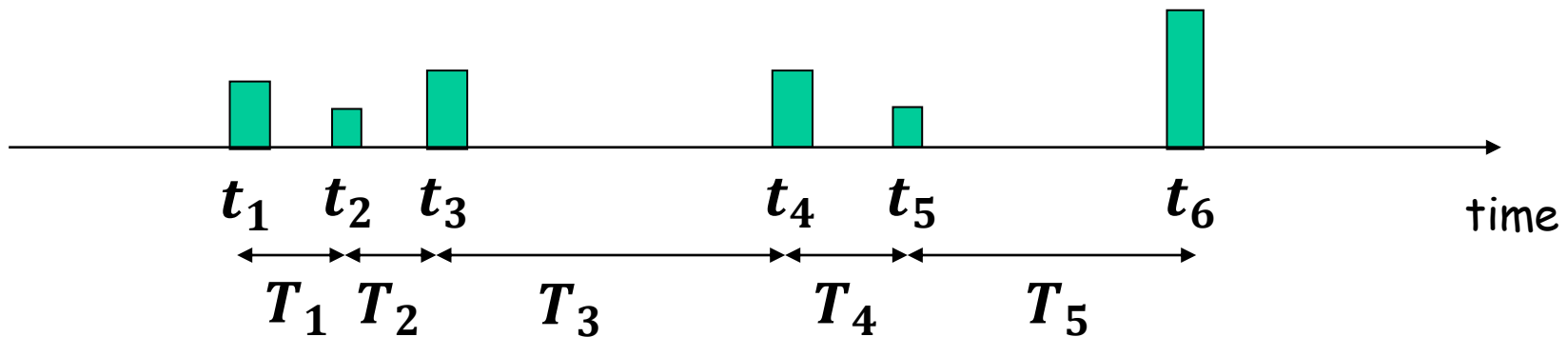
- ❖ starting time doesn't matter (memoryless prop.)
- ❖ merging 2 Poisson processes with rate λ_1 and λ_2
- creates a new Poisson process with $\lambda = \lambda_1 + \lambda_2$

Poisson and Merging Property



□ T is exponential with rate $\lambda = \lambda' + \lambda''$

Model the packet flow: Service Time



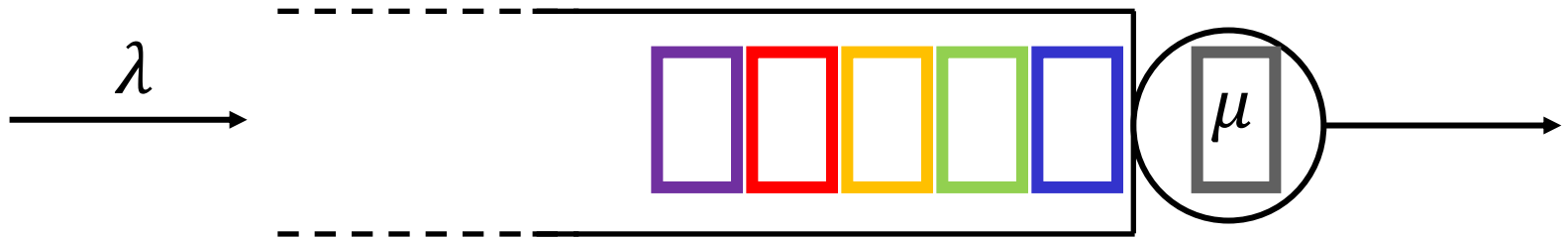
❑ Packets have various lengths

- ❖ takes different amount of time to process
 - the same packet given different link capacity/rate
 - packets with different lengths under a fixed link

❑ Service time S_i

- ❖ processing time of packet i under a fixed link rate
- ❖ follows i.i.d. r.v. S with mean $E[S] = 1/\mu$

M/M/1 Model



- ❑ Single server with a queue of infinite size
- ❑ Poisson arrival with rate λ
- ❑ Exponential i.i.d. service time with rate μ
- ❑ Arrival and service times are independent
- ❑ First-In-First-Out (FIFO) service discipline
- ❑ Check `mm1-console.html`

First Result

□ Utilization ρ is defined as

- ❖ percentage of time that the “server” is busy, or
- ❖ probability a random observation finds server busy
- ❖ a general result (applies but not limited to M/M/1)

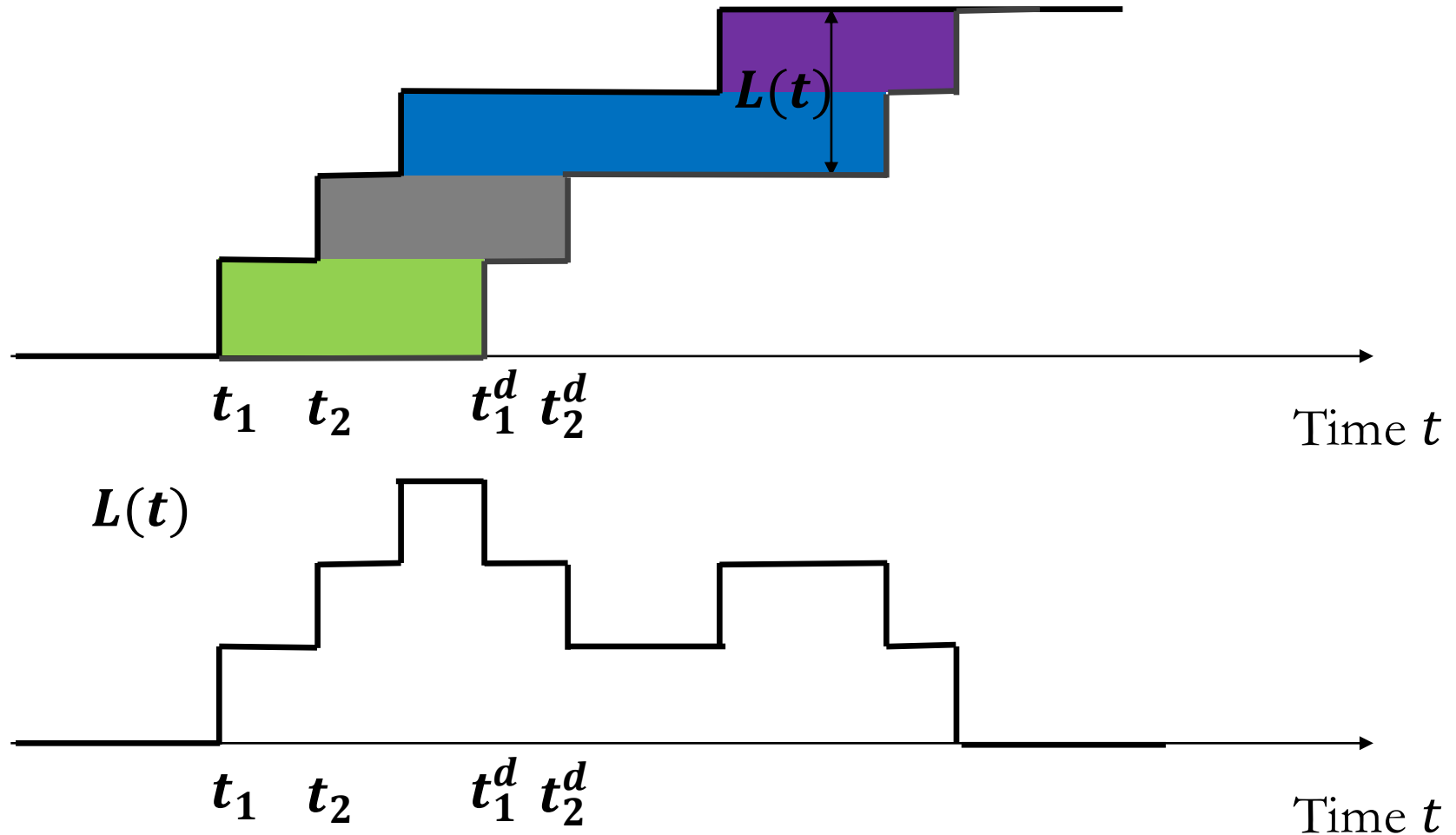
$$\rho = \frac{\lambda}{\mu}$$

- ❖ need a condition of $\lambda < \mu$ for system stability

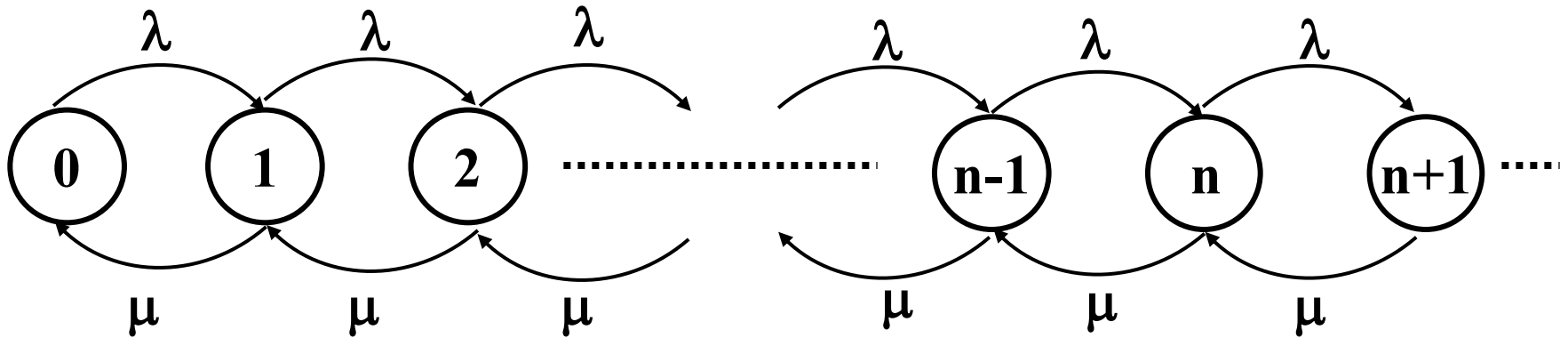
□ How can we understand this result?

□ Can we also derive it by using Little's Law?

$L(t)$ as the system state



Birth-Death Process



- ❑ State transition diagram for $M/M/1$ system
- ❑ Define the system state L as the (random) number of users in the system
- ❑ Transition to a new state: the rate of that event happening under the current state

Main Result (without proof)

□ Denote π_i as

- ❖ Percentage of time that exactly i packets or customers in the system, i.e., server + queue
- ❖ Also the probability $P\{L = i\}$ that a random observation finds i packets in the system
- ❖ For M/M/1 system, we have

$$\pi_i = P\{L = i\} = \rho^i(1 - \rho)$$

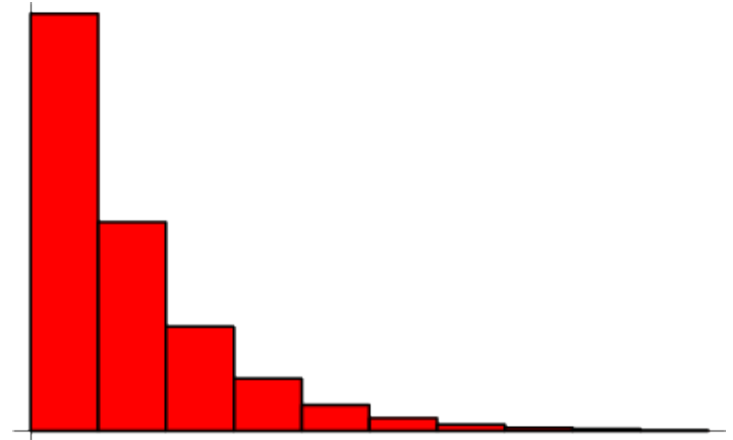
□ As a sanity check

- ❖ what is the value for π_0 ?
- ❖ what is the sum of all the π_i s?
- ❖ follows a geometric distribution

Geometric Distribution

- Discrete distribution

- ❖ take values $0, 1, \dots$
- ❖ density function looks like:
- ❖ $P\{L = i\} = p(1 - p)^i$



- Used to model the random number of coin flips of TAIL(s) until you get a HEAD ...
- Has some similarity with the exponential distribution

Now you know all about M/M/1

□ Look at the system, i.e., server + queue

- ❖ Average # of packets in the system

$$E[L] = \frac{\rho}{1 - \rho}$$

- ❖ Average sojourn time of packets

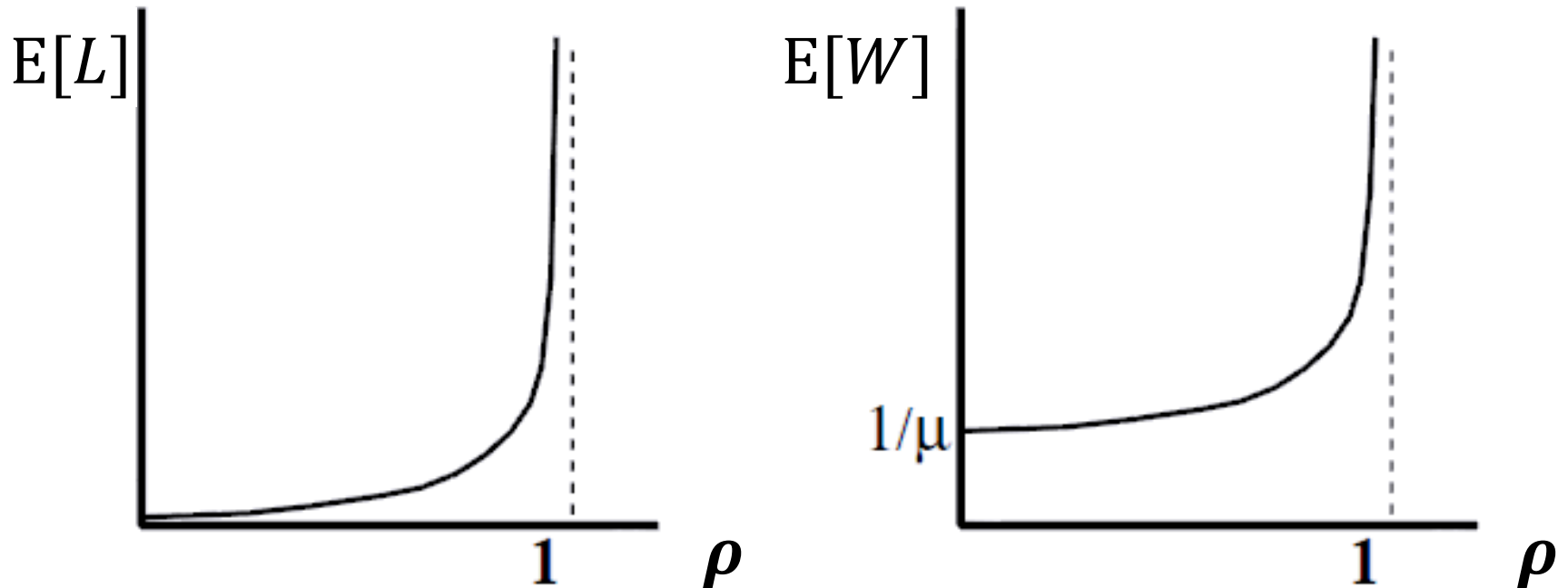
$$E[W] = \frac{1}{\mu - \lambda}$$

□ Focus on the queue

- ❖ Average # of packets in the queue $E[Q] = \frac{\rho^2}{1 - \rho}$

- ❖ Avg. queueing delay of packets $E[D] = E[W] - \frac{1}{\mu}$

Stability of M/M/1

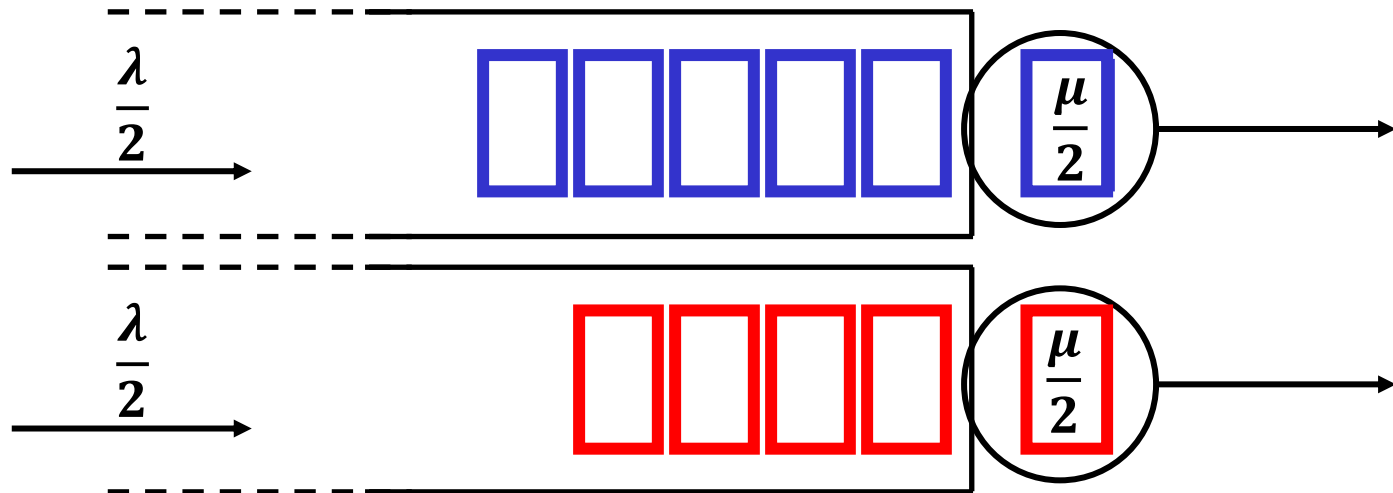
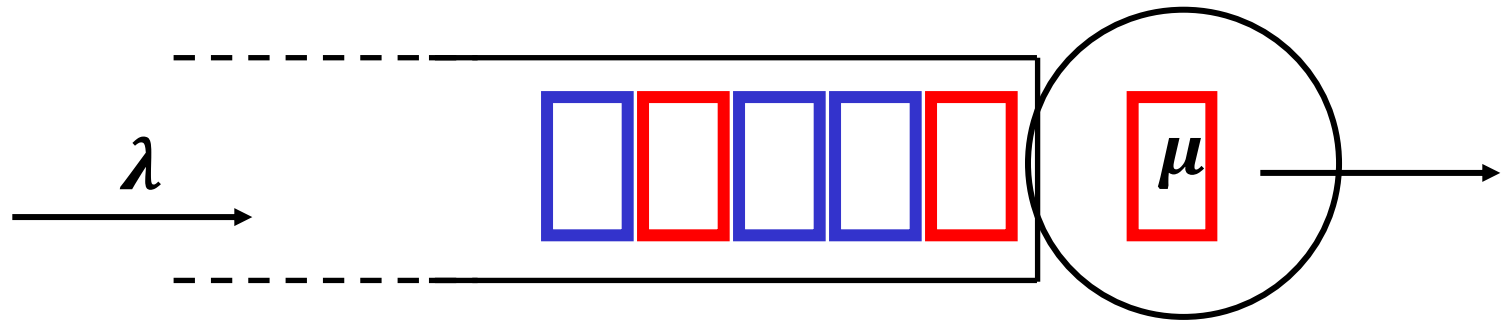


- This is why $\lambda = \mu$ is unstable
- What is the throughput of the system?

Effective Bandwidth

- Physical link capacity
 - ❖ e.g., 10 Mbps Cable
 - ❖ theoretical processing limit of the hardware
- Effective bandwidth of a link
 - ❖ the actual throughput that can be achieved
 - ❖ also depends on the quality of service achieved
- A physical capacity μ of an $M/M/1$ link
 - ❖ what is the maximum achievable throughput if
 - Utilization is below $\hat{\rho}$?
 - Average queueing delay is bounded by $\hat{D}$?

Statistical Multiplexing vs TDM



Statistical Multiplexing vs TDM

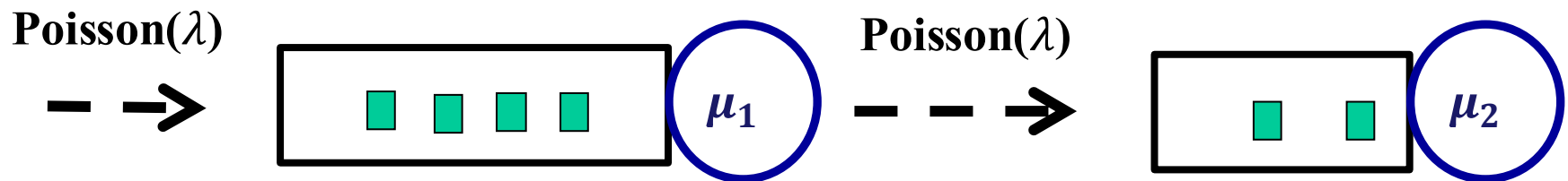
- ❑ Allocate each Poisson stream its own queue $(\frac{\lambda}{k}, \frac{\mu}{k})$ or shared a single queue (λ, μ) ?
- ❑ $\rho = \frac{\lambda}{\mu} = \frac{\lambda/k}{\mu/k} \Rightarrow$ the same server utilizations
- ❑ $E[W] = \frac{1}{\mu} \frac{1}{1-\rho} \Rightarrow$ avg. sojourn time for each user increased by k times under TDM
- ❑ $E[Q] = \frac{\rho^2}{1-\rho}, E[L] = \frac{\rho}{1-\rho} \Rightarrow$ avg. # of waiting and total users are also k times larger

Various queueing models

- ❑ Multiple servers
 - ❖ e.g., $M/M/2$
- ❑ Limited queue size
 - ❖ $M/M/1/K$
- ❑ Different service disciplines
 - ❖ e.g., $M/M/1/LIFO$
- ❑ General service and inter-arrival times
 - ❖ service times: $M/G/1$, $M/D/1$
 - ❖ arrival times: $G/M/1$
 - ❖ both: $G/G/1$

Burke's Theorem

- If an $M/M/1$ system with arrival rate λ starts in a steady state
 - ❖ the departure process is Poisson with rate λ
 - ❖ the number of customers in the system at any time t is *independent* of the sequence of departure times prior to time t



Tandem Queues



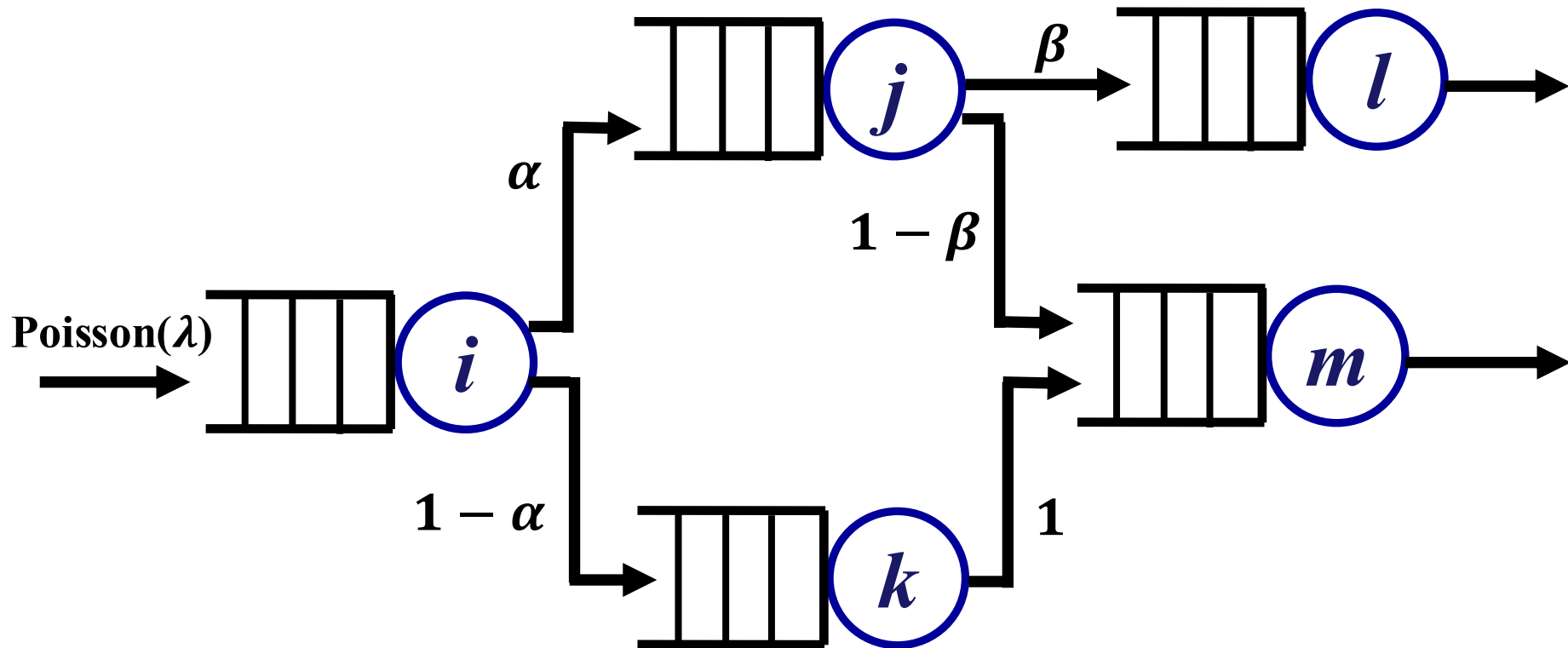
- the utilization of each server i becomes

$$\rho_i = \lambda / \mu_i$$

- by independence, the joint probability:

$$\begin{aligned} P\{L_1 = j, L_2 = k\} &= P\{L_1 = j\}P\{L_2 = k\} \\ &= \rho_1^j (1 - \rho_1) \rho_2^k (1 - \rho_2) \end{aligned}$$

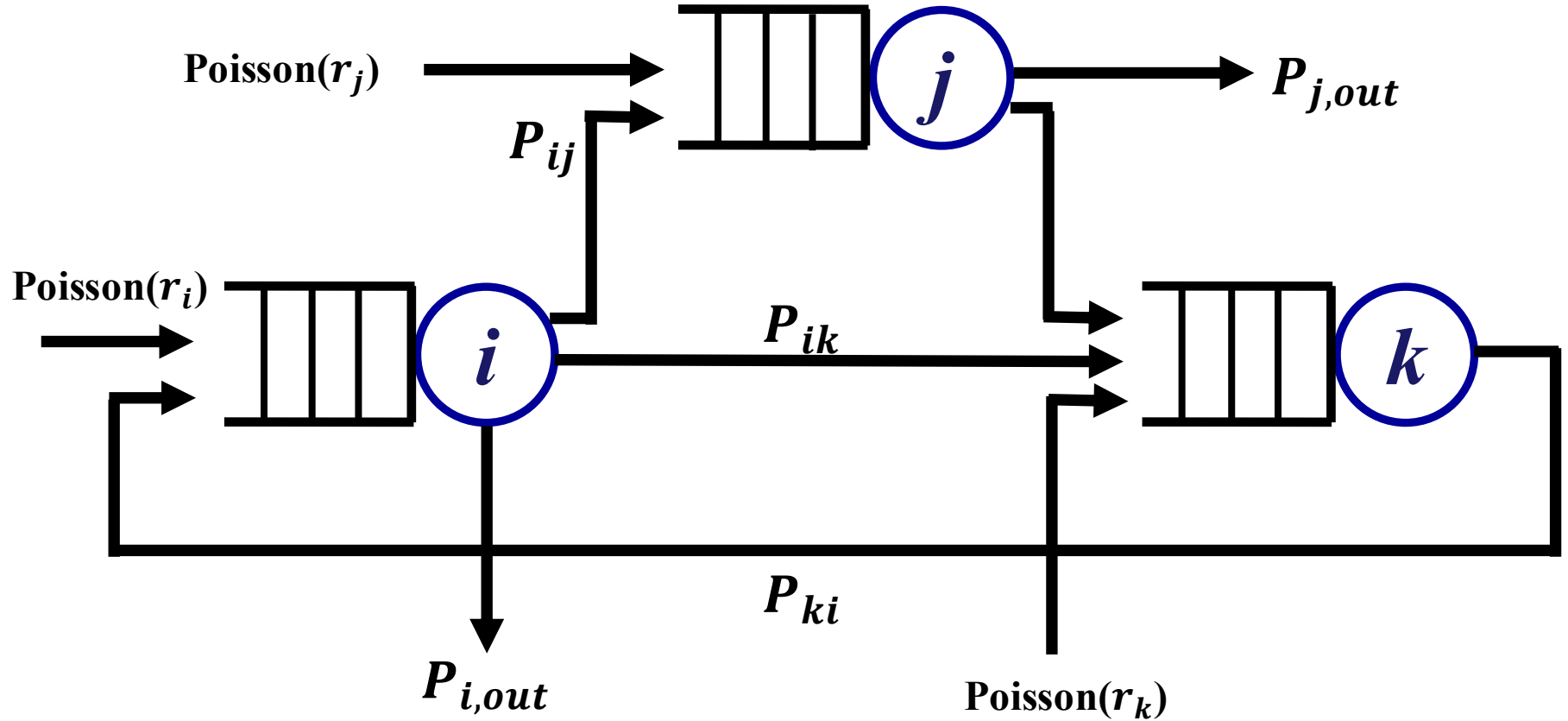
Acyclic Network with Probabilistic Routing



□ Capturing routing in a network

- ❖ Different flows might take different paths
- ❖ Each link is shared by different flows

Jackson Network



Solving the Jackson Network

- By Burke's theorem, merging and splitting of Poisson still apply

- A two-step procedure

 - ❖ Determine the utilization ρ_i for each server i

 - ❖ Product form solution as before, i.e.,

$$\mathbf{P}\{L_1 = l_1, \dots, L_n = l_n\} = \prod_{i=1}^n \mathbf{P}\{L_i = l_i\} = \prod_{i=1}^n \rho_i^{l_i} (1 - \rho_i)$$

- How to calculate ρ_i ?

 - ❖ How to calculate the effective arrival rate λ_i ?

Solving the Jackson Network

- The effective arrival rate λ_i include
 - ❖ Outside arrival directly to server i , i.e., r_i
 - ❖ Feedback arrivals from other servers

$$\lambda_i = r_i + \sum_{j=1}^n \lambda_j P_{ji}$$

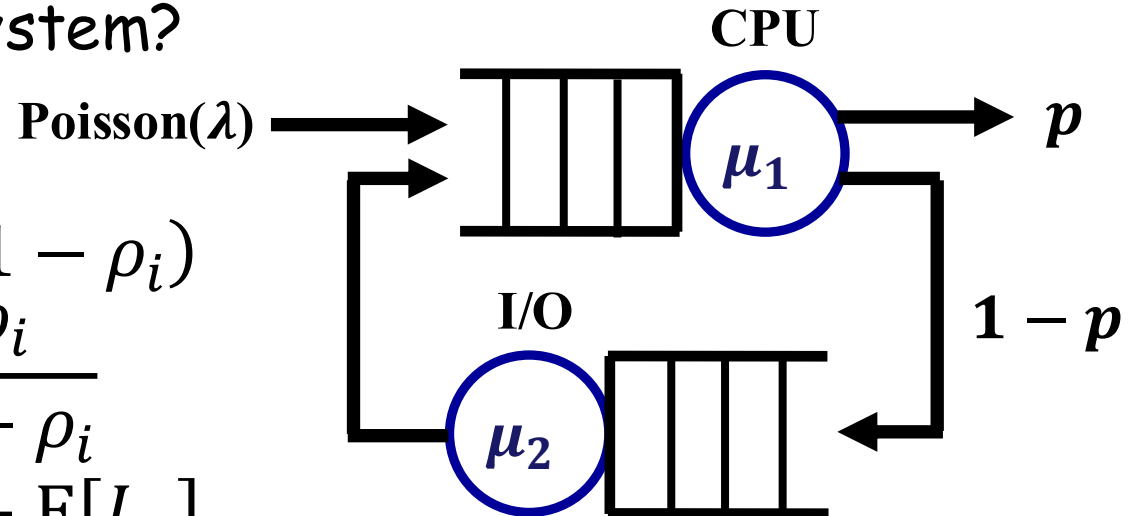
- ❖ In a matrix form

$$\lambda = r + \lambda P \Rightarrow \lambda = r(I - P)^{-1}$$

A simple web server example

- What is the average number of jobs in the system?

$$\begin{aligned}P\{L_i = x\} &= \rho_i^x (1 - \rho_i) \\ \Rightarrow E[L_i] &= \frac{\rho_i}{1 - \rho_i} \\ \Rightarrow E[L] &= E[L_1] + E[L_2]\end{aligned}$$



$$\begin{cases} \lambda_1 = \lambda + \lambda_2 \\ \lambda_2 = (1-p)\lambda_1 \end{cases} \Rightarrow \begin{cases} \lambda_1 = \frac{\lambda}{p} \\ \lambda_2 = \frac{(1-p)\lambda}{p} \end{cases} \Rightarrow \begin{cases} \rho_1 = \frac{\lambda}{p\mu_1} \\ \rho_2 = \frac{(1-p)\lambda}{p\mu_2} \end{cases}$$